Master DSC/MLDM - First year
Introduction to Artificial Intelligence
First exam - November 2024

Take the things you need with you. Leave your smartphone, calculator and computer in your bag and place your bag at the front of the examination room.
Time allocated: 3h - No documents allowed.
TAKE CARE: any cheating will be severely punished and will lead to a formal complaint to the disciplinary council of the university.

1 Truth table ($\simeq$ 2 points)

1. Using the truth table method, prove that the premises $(p \vee \neg q) \Rightarrow (r \wedge p)$ and $(r \vee q) \wedge \neg p$, logically entail $r \Rightarrow (q \vee \neg p)$.
2. Using the truth table method, prove that $(p \wedge (q \wedge r)) \Rightarrow ((p \wedge q) \Rightarrow (p \wedge r))$ is valid.

   For this second question, you should present your tables in the same way as the example below showing the truthtable of $(p \vee q) \wedge \neg p$:

p	q	(	p	∨	q	)	∧	¬	p
T	T		T	T	T		F	F	T
T	F		T	T	F		F	F	T
F	T		F	T	T		T	T	F
F	F		F	F	F		F	T	F

2 Problem solving using a truth table ($\simeq$ 2 points)

After an exhausting journey through the enchanted *Forest of Whispers*, three brave adventurers—*Mira*, *Leo*, and *Sam*—reach the entrance to the hidden *Temple of Elements*, where a legendary *Ring of Power* is said to be kept. However, before they can claim the ring, they face a challenge: they must choose between *three tunnels*. **One tunnel** leads to the ring, but **the other two** are traps, cursed with powerful magic that could imprison them forever.

Above each tunnel is a stone inscription, offering a cryptic clue:

1. (statement 1) The inscription above the **first tunnel** reads: "The ring is not in this tunnel."
2. (statement 2) The inscription above the **second tunnel** reads: "The ring is not in the first tunnel."
3. (statement 3) The inscription above the **third tunnel** reads: "The ring is in this tunnel."

The adventurers know two more things for sure:

1. **at least one** of the statements is true,
2. **at least one** of the statements is false.

You are asked to:

1. Choose 3 proposition constants wisely.
2. Write some proposition sentences that describe the constraints above.
3. Draw a truth table for those 3 constants and sentences.
4. Conclude which tunnel should Mira, Leo, and Sam choose to safely retrieve the *Ring of Power*.

3 Problem modeling and solving in PL ($\simeq$ 5 points)

Here are some details about a small world:

When the sun sets, the wolves howl. When the wolves howl, Alex cannot sleep. When Alex cannot sleep, he reads a book and makes tea. If Alex makes tea, Bella wakes up. When Bella wakes up, she gets angry and goes for a walk. If Bella goes for a walk and it is raining, she uses an umbrella. When Bella talks to her friend, she laughs and Alex reads a book. If Bella gets angry, she talks to her friend. When it rains, the garden gets flooded. When the garden gets flooded, Emma cancels her picnic. When Emma cancels her picnic, Bella is disappointed. If Bella uses an umbrella and is disappointeed, she cries. The sun sets. It is raining.

You are asked to:

1. Choose proposition constants wisely (list all your constants on your exam sheet).
2. Convert these English sentences into propositional logic sentences (list all those sentences on your exam sheet).
3. Convert these propositional logic sentences into CNF (list all your clauses on your exam sheet).
4. Use resolution reasoning in propositional logic to prove that, in this world: *"Bella laughs and cries"* (write all the steps of the reasoning on your exam sheet).

4 Validity, unsatisfiability, contingency ($\simeq$ 3 points)

Using resolution reasonning and the method studied during the course, say whether the following sentences are unsatisfiable, valid or contingent

1. $(p \wedge (q \vee r)) \Rightarrow ((p \Rightarrow q) \vee (p \Rightarrow r))$
2. $\forall X.\forall Y.(p(X,Y) \Rightarrow r(X)) \Rightarrow (\forall Y.\exists X.(p(X,Y) \wedge r(X)))$
3. $\forall X.(p(X) \Rightarrow \neg r(X)) \wedge \exists X.(p(X) \wedge r(X))$

Note: you must write down all the details of the conversion of each formula to CNF.

5 Resolution principle ($\simeq$ 3 points)

For each pair of clauses below, say whether the resolution principle can be applied. If yes, explain the steps you follow and give the resolvent(s), if no, explain why.

1. $\{\neg q(b,Y,g(b)), r(Y)\}$ and $\{p(X), q(X,a,g(X))\}$
2. $\{p(A,h(B),C), q(A,B)\}$ and $\{\neg p(X,h(Y),k), \neg r(W), \neg p(f(a,b),h(c),W)\}$
3. $\{p(X,a,Y), q(Y)\}$ and $\{r(X), \neg p(b,X,d)\}$
4. $\{p(A,b,b,F), q(A,F)\}$ and $\{\neg p(a,X,b,g(X)), r(X,Y), \neg p(a,d,Y,g(a)), q(Y)\}$
5. $\{q(T), r(X,Y,Z), \neg p(a,f(X,g(Y),Z),X,b,T)\}$ and $\{q(A,B), p(A,f(a,B,c),b,C,g(B)), r(C)\}$
6. $\{p(A), q(A,B), r(B), q(a,C), s(C), q(A,b)\}$ and $\{\neg q(A,b), \neg q(a,B), t(A), t(B)\}$

6 Problem modeling and solving in FOL ($\simeq$ 5 points)

Here are some informations about a simple world:

If a student S has some knowledge about a problem P and this student S has self-confidence, then this problem P is solvable. If a problem P is a known problem of a scientific domain D and a student S owns a magic wand, then the problem P is solvable. If a student S works on a course C and a problem P is studied in this course C, then that student S has some knowledge about this problem P. If a teacher T does a course C and a student S takes notes on this course C and appreciates the teacher T and learns the course C, then we can say this student S works on this course C. If a teacher T does a course C and a student S owns a smartphone and the course C is publicly available and the student S watches it, then we can say this student S works on this course C. If a student S captures a course C with his camera and converts it to a mp4 file F and put it on youtube, then this course C is publicly available. If a student S owns a paper and owns a pen, and attends a course C and publishes it on facebook, then this course C is publicly available. If a teacher T does a course C and this teacher T is in a room R and a student S is also in this room R, then the student S attends this course C. Smith does an AI course. Johnson does a ML course. Paul owns a paper, Paul owns a pen, Mary owns a smartphone, Harry owns a magic wand. Overfitting is a known problem of ML. Bias is a known problem of ML. Ethics is a known problem of AI. John has self-confidence. Mary captures the ML course with her camera. Mary converts the ML course to a mp4 file f1. Mary publishes the file f1 on youtube, Paul publishes the AI course on facebook. Mary watches the ML course. John takes notes on the AI course. John appreciates Smith. John learns the AI course. The problem of overfitting is studied in the ML course. The problem of ethics is studied in the AI course. Smith is in room L125. Paul is in room L125.

You are asked to:

1. Convert these English sentences into first-order logic sentences.
2. Convert these first-order logic sentences into CNF.
3. Use resolution reasoning in first-order logic, to get **one** answer to the question: *"Which problem is solvable?"*